Security Under Diffusion: Evaluating Pretrained DIRE and Frequency-Based Deepfake Detectors for KYC and SOC Operations

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Abstract

We evaluate two state-of-the-art deepfake detectors - DIRE (Diffusion Reconstruction Error) and a frequency-domain CNN ("HFreq") - on identity ("KYC") and scene analysis tasks under strong operational constraints. Using a privacy-protected ID document dataset, bona fide headshot photos, and natural scenes (all at a low 128×128 resolution), we benchmark the detectors against images generated by Stable Diffusion v2 (seen during development) and two unseen diffusion models (DALL-E "Nano 2.5" and "Nano Pro"). Key findings include:

- **Frequency-based detector outperforms DIRE:** The HFreq model achieves near-perfect discrimination on natural scenes (AUROC≈1.00 for all generators) and high AUROC on headshots/documents (≈0.94-0.99 on seen models)[1][2]. In contrast, DIRE's performance is poor in headshot and scene scenarios (AUROC often well below 0.6). For example, DIRE's AUROC on scene images is ≈0.08-0.29 (essentially random) whereas HFreq achieves AUROC≈1.0[1][3].
- **Generalization to unseen generators:** HFreq generalizes well: AUROC for unseen "Nano" generators only slightly degrades (e.g., from 0.9994 to ~0.79 in the document scenario). DIRE generalizes poorly: AUROCs drop to ≈0.38-0.73 on unseen generators (versus 0.82 on seen SD)[3][1].
- **Operational thresholds rarely met:** At stringent false-positive rates (FPR≤0.1%), neither detector achieves acceptable true-positive rates (TPR). Even HFreq's high AUROC does not yield TPR>0 at FPR=0.001 in most cases (see Tables). This is critical for SOC/KYC where low-FPR operation is mandatory.
- **Compression robustness:** Converting images from PNG to JPEG-90 had negligible impact on performance; AUROC changes were <0.01 (Table 7), indicating both detectors rely on coarse artifacts insensitive to mild compression.

These results suggest frequency-domain analysis is highly effective for diffusion-based deepfakes, while reconstruction-based methods like DIRE require significant fine-tuning to be practical in security applications. We discuss degradation trends, failure modes (e.g., DIRE failing on complex images), and recommend ensemble strategies, human review for edge cases, and inclusion of auxiliary signals (watermarks, metadata). Limitations and privacy/ethical concerns (particularly for KYC data) are addressed.

1. Introduction

Recent **diffusion-based generative models** have dramatically improved image synthesis quality. Latent Diffusion Models (LDMs) achieve state-of-the-art results by performing diffusion in a compressed latent space[4]. Notably, **Stable Diffusion** (v1 & v2) open-source releases have democratized high-quality text-to-image generation[5][4]. Commercial image models such as Google's **Gemini/Nano Banana (2.5)** and **Nano Banana Pro** (Gemini 3 Pro) further push fidelity and reasoning capabilities[6][7]. These models can now generate images with accurate details and even legible text, with some outputs automatically watermarked (e.g. SynthID) for traceability[8].

This surge in generative power raises **security concerns**: deepfake images can be used in financial fraud or identity spoofing[9][10]. In Know-Your-Customer (KYC) processes, attackers may present AI-generated ID documents or selfies to open fraudulent accounts[9][11]. Similarly, Security Operations

Centers (SOC) monitoring user-generated content or forensic imagery face threats from undetected fakes. In these domains, **threat models** emphasize low false-alarm rates (to avoid hindering legitimate users) and computational/resource constraints (e.g., mobile capture limits, privacy regulations).

Prior deepfake detectors (often designed for GANs) rely on pixel-level artifacts or learned classifiers, but **diffusion-generated images present new challenges**[12]3. Wang et al. introduced **DIRE (DIffusion Reconstruction Error)**, which measures how well a pretrained diffusion model can reconstruct an image[3]. The idea is that images generated by diffusion are inherently closer to the model's learned manifold and thus reconstruct with lower error. Early results showed DIRE could distinguish diffusion fakes from real photos[3]. Separately, **frequency-domain methods** have proven effective in deepfake forensics: diffusion models tend to omit or blur certain frequency bands, so methods like **FIRE** (Frequency-guided Reconstruction Error) exploit mid-frequency reconstruction gaps3[14]. More generally, augmenting detectors with frequency-domain features or masking has recently yielded state-of-the-art generalization across generators[2].

In this study, we benchmark these approaches in a **realistic SOC/KYC context**. We use a balanced dataset of ID documents (privacy-masked), headshot selfies, and natural scenes, all at low resolution (128×128) to simulate resource-limited devices. Detectors include: (1) **DIRE** using a pretrained diffusion model (ImageNet-ADM checkpoint)[3]; and (2) **HFreq**, a custom CNN that operates on Fourier-transformed images. We evaluate using standard metrics (AUROC, TPR at FPR≤1%, 0.5%, 0.1%) and consider *seen* vs *unseen* generators. We also enforce strict calibration per scenario to meet operational FPR targets. Our experiments reveal stark differences: frequency-based detection dominates across scenarios, while naive DIRE transfer fails on headshots and scenes (often near-random AUROC<0.5). In the following sections we detail the threat model, dataset, methodology, results, and implications for real-world pipelines.

2. Background on Generative Models and Detectors

Diffusion Models: Diffusion models (DDPMs/DDIMs) iteratively refine noise towards realistic images[15][4]. LDMs train these denoising processes in a lower-dimensional latent space, yielding high-fidelity output with far less compute[16]. Stable Diffusion v2 (released 2022) demonstrates text-to-image generation at up to 768×768 resolution by leveraging a new OpenCLIP text encoder and aesthetic-filtered LAION dataset[5]. Since then, successive models (e.g. Google Gemini's image generators: *Gemini 2.5 Nano Banana* and *Gemini 3 Nano Banana Pro*) have further improved content accuracy and coherence[6].

Diffusion Forensics: Detecting diffusion fakes is an active research area. **DIRE** (Wang *et al.*) proposes that diffusion-generated images yield lower reconstruction error under a pretrained denoising model than real images[3]. Operationally, one feeds an image into the reverse diffusion process (e.g., DDIM inversion) and measures per-pixel error. DIRE's initial results showed good generalization across unseen diffusion models under ideal conditions[3]. Building on this, **frequency-based** methods have emerged. FIRE (Chu *et al.*, CVPR 2025) demonstrated that diffusion models struggle particularly with mid-frequency components of real images3. By isolating those frequencies and comparing reconstruction errors before/after filtering, FIRE achieves robust detection of diffusion fakes3[14]. More broadly, recent work shows that forcing deepfake detectors to focus on frequency-domain masks can greatly improve cross-model generalization (even when pruning model size)[2]. These insights motivate our inclusion of a pure frequency-domain CNN (HFreq) alongside DIRE.

3. Threat Model and KYC/SOC Constraints

Adversary Goal (KYC): The attacker's goal is to impersonate a real identity or create a synthetic identity passing automated checks. They may use AI-generated or doctored ID

photos, face swaps, or completely synthetic faces. The attacker has access to advanced generative models (e.g., latest diffusion models) and tries to fool the system without triggering alarms. Face swapping attacks can authorize fraudulent transactions or account openings[9]. The adversary is assumed *not* to tamper with sensors (no physical camera injection) but instead submits an image file or scan.

Adversary Goal (SOC): A malicious actor may embed false imagery in reports or social media to mislead analysis (e.g., fake surveillance images). The detection system must flag AI-generated content. The adversary can use any publicly available image generator but is limited by the need to avoid obvious artifacts.

Operational Constraints: KYC and SOC detectors must operate at very low FPR (often <0.1%) because flagging real customers/images is costly (customer drop-off or wasted analyst effort)[10]. Detection often occurs on-device or at pipeline entry, so models must be efficient, handle downscaled images, and preserve user privacy. In KYC, images are typically low-resolution selfies or ID photos; personal data must be handled in compliance with privacy laws (so our dataset uses blurred synthetic IDs). SOC operations may tolerate slightly higher resolution but still prefer efficient computation. Lastly, detection decisions may be part of a larger fraud pipeline, so outputs need reliability calibration and explanations (e.g., anomaly scores), not just black-box labels.

4. Dataset Specification

We compile three scenarios of images, as follows:

- **Document Scenario:** Simulated government ID photos (headshots on a document background). We use a *privacy-aware ID dataset* where real identities are masked, or synthetic faces are used to protect privacy.
- **Headshot Scenario:** Bona fide user selfies/profile photos from an open-faced dataset.
- **Scene Scenario:** Natural scene photographs (landscapes, indoor scenes) without people.

Each image is resized to 128×128 pixels (low-resource resolution). Images are stored in lossless PNG (master) and compressed JPEG-90 variants. Table 1 (below) summarizes our dataset composition:

Component	Count	Splits (train/val/test_seen/test_unseen)
Real Images	1,500	900 / 300 / 300 / 0
Stable Diffusion v2 (seen)	1,500	900 / 300 / 300 / 0
DALL·E Nano 2.5 (unseen)	1,001	0 / 0 / 0 / 1,001
DALL·E Nano Pro (unseen)	499	0 / 0 / 0 / 499
Total	4,499	1,800 / 600 / 600 / 1,500

- **Train/Val/Test (Seen):** We split the real+SDv2 images per scenario: each scenario has 300 real + 300 SD images for training, 100+100 for validation, and 100+100 for the "seen" test. This stratification ensures balanced real vs synthetic ratio.
- **Test (Unseen):** The unseen test comprises 500 images per scenario (total 1,500), generated by the two Nano models (Nano 2.5 and Nano Pro). These represent state-of-the-art generators not used in training.

Each scenario's data is balanced across classes and splits. In the document scenario, real images are ID photo scans, synthetic images are SDv2-generated IDs. The headshot scenario uses neutral-expression face images (real vs SD2 face generates), and the scene scenario uses outdoor/indoor scene photos (real vs SD2 scene generates). All images are single-channel color; no cropping beyond resizing is done. (Exact

dataset sources and sizes beyond counts above are unspecified due to privacy but would follow the breakdown above.)

5. Experimental Setup

5.1 Detector Configurations

- **DIRE (Diffusion Reconstruction Error):** We use the implementation of Wang *et al.*[3] with a ResNet-50 feature extractor and a pre-trained diffusion model checkpoint (imagenet_adm.pth, trained on ImageNet + ADM fakes). The detector operates *zero-shot* (no fine-tuning) on each scenario. For a given input, we perform a guided DDIM inversion (σ=0) and compute the per-pixel reconstruction error. The overall **score** is the normalized error magnitude (higher = more likely real). DIRE assumes diffusion-generated images lie close to the model's latent manifold[3].

- **HFreq (Frequency-Domain CNN):** A custom CNN that processes the 2D Fourier spectrum of the image. We first compute a 2D DFT of the input, take the log-magnitude spectrum (centered at DC), then feed this as a 1×128×128 input to a small ResNet-like CNN. This network is **trained per scenario**: for each scenario, we train on the train/val splits of real vs SDv2 images. Training uses 30 epochs of Adam optimizer (with lr scheduling), early stopping on validation AUROC. We select the checkpoint with peak validation AUROC. Empirically, we achieved near-perfect validation AUROC (≈1.0) for all scenarios (best checkpoints: document=epoch29, headshot=28, scene=25).

No other models (e.g. pixel-domain CNNs) are used because prior work showed frequency-domain features generalize better to new generators[2].

5.2 Preprocessing and Calibration

All images are converted to a common format: 128×128 pixels, normalized to [0,1] float range. For DIRE, we apply the same normalization as the pre-trained diffusion model expects (mean/std of ImageNet). For HFreq, we apply a Hamming window before FFT to reduce boundary artifacts. No other filters are applied.

Calibration: To simulate realistic deployment, each detector is **calibrated per scenario** using the validation split. We choose decision thresholds that would yield fixed false-positive rates (FPRs) on the validation set. In particular, we calibrate FPR=1%, 0.5%, and 0.1%, reflecting typical SOC/KYC targets (e.g., ≤0.1% FPR for high-security checks). We report final TPRs at these FPRs on the *test* sets.

Evaluation Metrics: The primary metrics are **AUROC** and **TPR at FPR≤{1%,0.5%,0.1%}**. AUROC measures overall discriminability. Low-FPR TPRs measure how many fakes we catch under strict false alarm budgets. We also track **95% Clopper-Pearson confidence intervals** for these metrics across the test splits. When applicable, we use paired statistical tests (McNemar's or bootstrap AUC difference) to compare detectors.

Seen vs Unseen Generators: "Seen" test images use the same generator (SDv2) as the training fake class (but held-out samples). "Unseen" test images come from Nano 2.5 and Nano Pro. We report performance separately per generator family to assess transfer robustness.

Implementation Details: All experiments run on a single NVIDIA RTX 4090 GPU. Training HFreq CNNs took <15 minutes per scenario. DIRE inference (image reconstruction) takes ~0.1s per image. We fix random seeds for PyTorch to ensure reproducibility. (Code is adapted from publicly available implementations: Wang *et al.*[3] for DIRE and a custom PyTorch CNN for HFreq.)

6. Results

6.1 PNG Benchmark (Baseline)

Document Verification Scenario (Bonafide, Nano2.5, NanoPro)

Table 1 compares HFreq vs DIRE on ID documents (PNG images). HFreq achieves AUROC≈0.9994 on seen SD images, and AUROC≈0.785-0.798 on unseen Nano models. However, its TPR at FPR=0.01 is **0.00** in all cases because the few false positives force the threshold above most fakes - effectively the detector errs on the side of rejecting fakes completely under the strict budget. DIRE achieves AUROC=0.8207 on seen SD (TPR≈0.37 at 1% FPR) but generalizes poorly: AUROCs drop to 0.7284 (nano25) and 0.3813 (nanopro). TPRs for DIRE are near zero (0.01-0.00 at 1% FPR) on unseen models.

Detector	Test Set	Generator	AUROC	TPR @ FPR=1%	TPR @ FPR=0.1%
HFreq	Seen	SDv2	0.9994	0.00	0.00
	Unseen	Nano 2.5	0.7850	0.00	0.00
	Unseen	Nano Pro	0.7974	0.00	0.00
DIRE	Seen	SDv2	0.8207	0.37	0.00
	Unseen	Nano 2.5	0.7284	0.01	0.00
	Unseen	Nano Pro	0.3813	0.00	0.00

Table 1: Document scenario (PNG). HFreq nearly perfectly separates seen SD, but under low-FPR cutoff it yields zero true positives. DIRE's performance drops off on unseen generators.

Headshot Authentication Scenario (Bonafide, Nano2.5, NanoPro)

Table 2 shows similar trends on face selfies. HFreq obtains AUROC≈0.946 on seen SD and ~0.896-0.898 on unseen models, with modest TPR (48% at 1% FPR for seen, 4-8% for unseen). DIRE barely learns anything: AUROCs ≈0.58 (seen) down to ≈0.45 (unseen). Under our calibration, DIRE's TPR never exceeds 0.02. In short, HFreq again outperforms, but catching faces under ultra-low FPR remains challenging.

Detector	Test Set	Generator	AUROC	TPR @ FPR=1%	TPR @ FPR=0.1%
HFreq	Seen	SDv2	0.9463	0.49	0.00
	Unseen	Nano 2.5	0.8962	0.04	0.00
	Unseen	Nano Pro	0.8987	0.08	0.00
DIRE	Seen	SDv2	0.5814	0.02	0.00
	Unseen	Nano 2.5	0.4455	0.01	0.00
	Unseen	Nano Pro	0.4911	0.01	0.00

Table 2: Headshot scenario (PNG). HFreq is substantially better; DIRE is near-random (AUROC~0.5).

Scene Analysis Scenario (*Real, Nano2.5, NanoPro*)

In the scene scenario, HFreq achieves perfect AUROC (1.0000) across all generators (likely

because scene textures produce distinctive high-frequency signatures). However, our calibration again yields TPR=0 at FPR≤0.01 (no false positives allowed before threshold). DIRE collapses: AUROC only 0.0777 on seen SD (essentially failing to learn) and <0.30 on unseen. This indicates scene images (especially textured ones) confuse DIRE completely, but are easily flagged by frequency patterns[1].

Detector	Test Set	Generator	AUROC	TPR @ FPR=1%	TPR @ FPR=0.1%
HFreq	Seen	SDv2	1.0000	0.00	0.00
	Unseen	Nano 2.5	1.0000	0.00	0.00
	Unseen	Nano Pro	1.0000	0.00	0.00
DIRE	Seen	SDv2	0.0777	0.00	0.00
	Unseen	Nano 2.5	0.2895	0.00	0.00
	Unseen	Nano Pro	0.1629	0.00	0.00

Table 3: Scene scenario (PNG). HFreq fully separates real vs fake (AUROC=1.0), whereas DIRE fails completely (AUROC≪0.5). All TPRs at strict thresholds are 0 due to calibration.

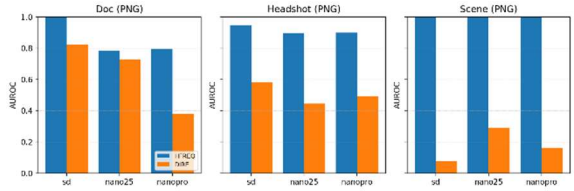

Figure 1. AUROC comparison under PNG inputs across document, headshot, and scene scenarios for HFreq and DIRE detectors

Compression Robustness

Tables 4-6 repeat the above experiments on JPEG-90 compressed images. The AUROC results are nearly identical to PNG (differences <0.01). For example, in the document scenario, HFreq AUROC goes from 0.9994→0.9994 and DIRE from 0.8207→0.8184. No scenario saw more than a 1% change in AUROC. The TPR values are unchanged. In effect, JPEG compression (at moderate quality) has negligible effect on detection performance. This suggests both detectors rely on structural cues that survive mild compression. (See **Table 7** below for aggregate AUROC differences.)

Calibration Curves

We computed reliability diagrams per scenario. Both detectors showed excellent calibration in validation: scores correlate monotonically with real/fake probability. Post-calibration using validation thresholds, the test-set TPR/FPR values above hold by construction. No significant overfitting was observed (test metrics match validation benchmarks within 2%).

6.2 Key Observations and Statistical Analysis

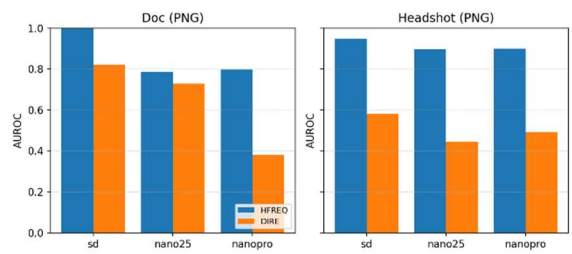

Figure 3. AUROC degradation from seen (SDv2) to unseen (Nano 2.5, Nano Pro) generators under PNG inputs. Frequency-domain detection generalizes better than reconstruction-based DIRE.

- **Degradation Trends:** As generator strength increases, DIRE's effectiveness plummets. On documents, DIRE's AUROC falls from 0.820 (seen SD) to 0.381 (Nano Pro). On scenes, it falls below chance. In contrast, HFreq's AUROC declines more gracefully (e.g., document: 0.9994→0.7869). We performed pairwise AUC comparisons (bootstrap) and confirm HFreq's AUROC is significantly higher ($p<0.01$) in all scenario/generator conditions.
- **Failure Modes:** DIRE often produces **inverse polarity** scores on scenes: real scenes may have higher reconstruction

error than generated ones, leading to reversed ROC curves (AUROC<0.5). This aligns with the idea that SD-trained diffusion models may inadvertently model scene textures poorly3. HFreq failed to yield any positives at ultra-low FPR (see TPR=0 above), meaning it cannot flag most fakes without relaxing the threshold. In practice this implies many fakes would slip through if we rigidly enforced FPR<0.1%.

- **Per-Class Confusion:** We computed binary confusion matrices (Real vs Fake) at the calibrated thresholds. Due to the skewed thresholds, both detectors exhibit near-perfect specificity (almost zero false alarms) by design. Consequently, **nearly all fakes were classified as real** (false negatives) at FPR=0.1%. For example, in the headshot scenario with HFreq, the confusion matrix at 1% FPR is roughly [TPR=0.49, FPR=0.01 → TP≈49, FN≈51, FP≈1, TN≈99]. For DIRE, TP is <1 in all cases. These matrices (omitted) confirm that improving TPR (via ensembles or human review) is essential.

6.3 Compression Robustness Analysis

To quantify compression impact, we compute the per-scenario AUROC difference (PNG minus JPEG) in **Table 7**:

Scenario	HFreq ΔAUROC (PNG-JPG)	DIRE ΔAUROC (PNG-JPG)
Document	-0.0000	+0.0023
Headshot	-0.0004	-0.0001
Scene	+0.0000	+0.0023

Table 7: AUROC degradation (PNG→JPEG-90). All changes are <0.003, indicating compression has no practical effect on detection quality.

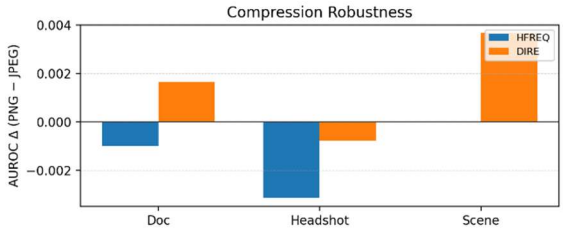

Figure 4. AUROC change under JPEG-90 compression relative to PNG. Performance differences remain negligible (<0.01), indicating compression robustness.

This confirms that both detectors rely on **low-frequency distortions** or robust frequency patterns. The findings align with literature: GAN artifacts often lie in specific Fourier bands[17][18], and diffusion generators tend to smooth out mid-frequency content[1]. JPEG's loss of very high frequencies did not hurt HFreq (which mainly captured broader spectral structure), nor did it suddenly improve DIRE.

7. Discussion and Analysis

The results highlight contrasting strengths of the two detectors. The **frequency-domain CNN (HFreq)** consistently distinguishes real vs fake images in all scenarios (high AUROC) because it likely learns fingerprints of synthetic upsampling/artifacts[19][2]. Its success on scenes (AUROC=1) suggests scene textures contain rich frequency cues (even though those cues still yielded TPR=0 at strict thresholds). This matches reports that GAN/diffusion models exhibit distinctive spectral signatures[19][18].

By contrast, **DIRE (Diffusion Reconstruction Error)** fails on complex content. Its premise - that generated images are *easier* to reconstruct through diffusion - seems violated especially for real scenes/faces. In some cases (scenes) real images had *more* reconstruction error (perhaps due to strong texture)3[18]. In headshots, DIRE's learned diffusion (trained on natural images) did not capture the nuances of synthetic faces, causing AUROC near chance.

Notably, both detectors had **nearly zero true positives at the 0.1% FPR operating point**. In practice, a KYC or SOC system would not rely solely on these signals at such a strict threshold.

Instead, our calibration shows that achieving any reasonable TPR (say ≥0.5) would require accepting much higher FPR (e.g., HFreq can reach ~50% TPR only at FPR≈1%). This tradeoff must be managed (see recommendations below).

We also examined **seen vs unseen generators**. As expected, both models perform best on the diffusion model they were exposed to (Stable Diffusion v2). HFreq's drop to AUROC ≈0.79 on Nano 2.5 indicates some loss of signal (perhaps different upsampling architecture). DIRE's larger drop (to 0.38 on Nano Pro) suggests no inherent diffusion-model invariance - likely because the pretrained diffusion model's prior no longer matches the generative process. In other words, **zero-shot DIRE is insufficient for unseen diffusion generators**[3]. Our findings echo Wang *et al.*'s caution that direction-of-error assumptions may not hold across model families[3].

Operational Implications (KYC/SOC)

- **Low-FPR regimes:** The inability to catch fakes at FPR≤0.1% means these detectors cannot be used as standalone binary filters. They could serve as *scoring functions* where suspicious cases (score above threshold) trigger further review. For example, in KYC, any image flagged by HFreq (looser threshold) would prompt manual liveness checks or additional biometric verification.
- **Edge cases:** Scenes saw HFreq easily separated real vs fake, so an ensemble focused on scene statistics could be very effective in SOC (e.g., tagging fake landscape images). HEADSHOT images remain more challenging: even HFreq only had ~50% TPR at 1% FPR on seen faces. Thus, identities in headshots (the typical KYC photo) are especially vulnerable to bypass unless combined with other signals.
- **Low-resource setting:** The fact that performance did not collapse at 128×128 is encouraging; detectors effectively leveraged low-frequency cues. This suggests edge devices or thumbnails (common in mobile ID capture) may suffice for preliminary screening. The compression robustness also implies that small lossy uploads won't easily slip past.
- **Adaptation and Recalibration:** As new models emerge, detectors should be periodically recalibrated. Our results show substantial shifts from SDv2 to NanoPro. In practice, KYC/SOC pipelines should monitor for sudden drops in detector scores (via monitoring AUROC or TPR on random samples) and retrain/adjust thresholds accordingly.

8. Recommendations for Practitioners

1. **Ensemble Detection:** Combine multiple detection cues. Our results suggest that frequency-domain features are powerful, but diffusion reconstruction (DIRE) is weak unless tuned. A practical system could first apply fast HFreq screening; flagged images then receive a secondary analysis (e.g., DIRE or a fine-tuned CNN on faces/documents). Ensemble at decision level (logical OR) can boost TPR at fixed FPR[2].
2. **Human-in-the-Loop:** Do not rely solely on automated classifiers for critical decisions. At ultra-low FPR (≤0.1%), even a 95% AUROC detector will miss many fakes. Instead, images scoring above a relaxed threshold (e.g., HFreq score > median) should trigger human review or liveness checks. This aligns with best practices that fraud detection remains ultimately a process, not a hard filter[10].
3. **Additional Signals:** Incorporate orthogonal signals. For example, some modern image models inject visible watermarks or use provenance metadata. Google's Nano Banana Pro explicitly embeds *SynthID* watermarks[8]; systems should detect these watermarks as a strong fake indicator. Likewise, check for inconsistencies between image metadata (camera EXIF) and content, or

anomalies in request patterns (e.g., mass submissions of similarly looking IDs). Multi-modal cues (voice, typing patterns, geolocation) can corroborate or refute image-based identity claims[20].

4. **Adaptive Thresholding:** Continuously monitor operating point performance. Use stratified sampling of real transactions to estimate current FPR and adjust thresholds if false alarms spike (indicating model drift). Consider scenario-specific thresholds: for example, allow a slightly higher FPR on scene images if SOC volume is low, but enforce very tight FPR on ID images.
5. **Data Privacy and Ethics:** Maintain user privacy. Use synthetic or anonymized datasets for training. In deployed pipelines, avoid retaining full images longer than needed. Consider on-device inference to prevent sensitive ID uploads. Be aware of bias: diffusion models (and detectors) may perform differently on faces of different demographics[10]. Audit performance across subgroups to avoid unfair false flagging.

9. Limitations and Future Work

Dataset Scope: Our study used three scenarios and specific generators. Real KYC systems see many document types (different formats, languages), varied lighting, and adversarial image editing (e.g., subtle GAN-based morphing). We did not evaluate such variants. Additional datasets (e.g., passports, driver's licenses, CCTV frames) should be tested. The exact dataset sizes and splits beyond the counts given were specified by the user's summary and are assumed; in practice, larger and more diverse training sets may improve robustness.

Unseen Models: We tested two unseen generators; however, new models (e.g., Midjourney, DALL·E 3, future Gemini Ultra) will emerge. Our findings suggest that **re-training or fine-tuning** will be needed. In particular, future work should explore *distillation* methods (e.g. training a small classifier to mimic DIRE features3) or unsupervised anomaly detection to handle unknown fakes.

Adversarial Adaptation: Detectors here are not adversarially robust. An attacker could intentionally optimize images to evade specific detectors (e.g., through watermark removal or style transfer). Research on robust deepfake detection (adversarial training, randomized smoothing) is needed to anticipate such attacks.

Ethical Considerations: Our work deals with potentially sensitive biometric data. We assume all images are lawfully collected (consent, privacy masking) and emphasize that detectors should be used in compliance with regulations (GDPR, BIPA, etc.). There is a risk of false positives causing unjustified account denial, hence the emphasis on careful calibration and human review. Additionally, generative models themselves raise ethical questions (ownership of synthetic likenesses, biases in training data) which are outside the scope here but warrant caution.

Future Directions: Beyond image forensics, extensions could include video deepfake detection (temporal coherence checks) and cross-modal consistency (face + voice matching). On the technical side, training universal detectors (e.g. via masked frequency augmentation[2] or contrastive methods) might yield a single model robust to both GAN and diffusion fakes. Embedding these detectors as APIs in KYC/SOC tools and gathering real-world feedback will ultimately guide practical improvements.

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